

Are indicative conditionals material conditionals?

Topics in Mind and Language: Conditionals

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1. The material conditional theory of indicative conditionals

An indicative conditional of the form 'If ϕ , ψ ' is true at all and only those possible worlds where ψ is true or ϕ is not true.

- We can get this as a limiting case from the variably strict theory by taking the accessibility relation to be one that every world bears to itself and no other world.

2. Reasons for thinking 'If P, Q' entails ' $P \supset Q$ '

This follows (given classical logic) from the validity of Modus Ponens for the indicative conditional. $P \supset Q$ and P together imply Q, so $P \supset Q$ is inconsistent with $P \wedge \neg Q$, so $P \supset Q$ implies $\neg(P \wedge \neg Q)$.

3. Arguments for the material conditional theory

The Or-to-if inference: 'P or Q; therefore if not-P then Q'.

- Seems valid! 'It's either a dog or a cat; so if it's not a cat it's a dog'.
- Or-to-if is valid if the indicative conditional is like $\supset$ in classical logic in obeying the metarule of *conditional proof*: whenever $P_1, \dots, P_n \vdash Q$, $P_1, \dots, P_{n-1} \vdash P_n \supset Q$

$P \vee Q, \neg P \vdash Q$	(disjunctive syllogism)
$P \vee Q \vdash \neg P \supset Q$	(conditional proof)

- Gibbard's proof: Or-to-if can be derived from uncontroversial principles of conditional logic together with the *Import-export* principle, according to which 'If A, then if B then C' is equivalent to 'If A and B, then C'. (Actually we only need the right-to-left, 'import' direction of this.).

$\vdash ((P \vee Q) \wedge \neg P) \supset ((P \vee Q) \wedge \neg P)$	(ID)
$\vdash ((P \vee Q) \wedge \neg P) \supset Q$	Weakening the consequent
$\vdash (P \vee Q) \supset (\neg P \supset Q)$	(Import-Export)
$P \vee Q \vdash \neg P \supset Q$	(MP)

A worry about this argument: Import-Export feels valid for counterfactuals too, but Or-to-if seems ridiculous for them!

4. A challenge to the material conditional theory:

Objection: asserting 'If P, Q' in a situation where you just know not-P, or just know Q, seems very bad and misleading. But according to the material conditional theory, you know it's true in this kind of case.

5. An analogy

According to orthodoxy, 'P or Q' is entailed by P and entailed by Q. One might object to the orthodoxy on the grounds that when you just know one of P and Q and have no evidence about the other, asserting 'P or Q' would be very bad and misleading, even though according to orthodoxy you're in a position to know that it's true.

Misleading how? Saying 'P or Q' will generally lead hearers to conclude that you don't know either way.

Response: a sentence can be misleading and bad to assert even if it's true.

- Strategic point: because of this obvious and pervasive phenomenon, objections of the form 'This theory should be rejected because it lacks a good explanation of why sentence S is bad to assert' are hard to make stick. The other phenomenon — something that's good to assert although false — is much rarer. So it might be easier to object to the material conditional theory on the grounds that it says certain things are false (or likely to be false) that are in fact good to assert.
- However, 'It is not the case that if P, Q' is pretty stilted so the fact that materialism makes it so hard for this to be true is not immediately terrible. But maybe we can do better by considering negative answers to conditional questions?